



Sensitivity Coefficients of a Single Response Function in Fixed-Source Calculations Using SERPENT2 and MCNP6

Yurdunaz Celik^{1,*}, Juan A. Monleon de la Lluvia², Gert van den Eynde¹

¹SCK CEN, Belgian nuclear research centre, Boeretang 190, 2400 Mol, Belgium;

²Autorité de Sécurité Nucléaire et de Radioprotection PSN-RES/SNC/LN, Fontenay-aux-Roses 92260, France

DOI: 10.5281/zenodo.20803819

ABSTRACT

Accurate evaluation of reaction rate uncertainties due to the nuclear data is essential for validating neutron transport simulations against experimental measurements. While the MCNP6 code can provide sensitivity information through its perturbation capabilities, it does not directly calculate the first-order sensitivity coefficients for fixed sources. The SERPENT2 code, on the other hand, lacks a built-in feature to compute sensitivities for individual reaction rates, as its implementation is inherently restricted to ratios of detector responses rather than single-response quantities. Moreover, while MCNP6 is supported by extensive documentation, SERPENT2 offers only limited guidance, particularly on the underlying mathematical formulations.

In this work, sensitivity coefficients for a selected reaction rate were calculated using both MCNP6 and SERPENT2 under fixed-source conditions. Another aim of this work is to establish a theoretical link between the internal formulations of the codes. By explicitly clarifying this link and demonstrating the workaround approach, this study demonstrates the feasibility of performing a reaction-rate sensitivity analysis in SERPENT2 and provides a reference for future uncertainty propagation studies of reaction rate calculations outside the core barrel in the VENUS-F reactor.

Keywords: Sensitivity coefficient, MCNP6, SERPENT2, Reaction Rate

1. INTRODUCTION

The VENUS-F reactor [1], located at the Belgian Nuclear Research Centre (SCK CEN), is a zero-power fast-spectrum critical facility using lead as a coolant. This configuration enables it to support research related to advanced fast reactor systems, particularly lead-cooled fast reactors (LFRs) and small modular reactors (SMRs) that operate in a similar fast neutron spectrum environment. The fast spectrum environment of VENUS-F provides a unique opportunity to experimentally study neutron transport and nuclear data effects under conditions representative of lead-fast cooled SMR designs.

Understanding the neutron spectrum outside the reactor vessel is essential, as it plays a key role in determining activation and shielding requirements. Accurate modeling of these effects requires reliable nuclear data and understanding the uncertainties in the calculated reaction rates. Consequently, it is important to evaluate how uncertainties in neutron cross sections propagate to uncertainties in reaction rate calculations, particularly for experimental validation campaigns that are planned in the future at VENUS-F.

The present work focuses on comparing the sensitivity coefficients obtained with MCNP6 [2] and SERPENT2 [3] codes under fixed-source conditions. While MCNP6 provides perturbation capabilities that can be used to derive first-order sensitivity coefficients, SERPENT2 restricts its sensitivity implementation to ratios of detector responses, preventing the direct evaluation of single-response sensitivities. This limitation is particularly relevant for fixed-source studies, where individual reaction-rate sensitivities are required. The main objective of this work is therefore to demonstrate and validate a practical approach that enables extraction of first-order sensitivity coefficients for a single reaction

rate in SERPENT2. Ultimately, this study aims to establish a reference approach for future uncertainty propagation studies of reaction rates measured outside the core barrel in the VENUS-F reactor.

2. METHODOLOGY in MCNP6

MCNP6 [2] offers a differential operator approach (PERT card) to perturbation theory, following the established first-order Taylor expansion framework described in Ref. [3]. The underlying perturbation formalism is briefly summarized here for completeness. MCNP6 uses a first-order (linear) and second-order (quadratic) Taylor expansion to approximate the new response (R) due to perturbations in input parameters, such as material density, material weight fractions, and geometry. The Taylor series expansion of a response that is a function of some reaction cross section σ is

$$R(\sigma_x) = R(\sigma_{x,0}) + \frac{dR}{d\sigma_x} \Big|_{\sigma_{x,0}} \Delta\sigma_x + \frac{1}{2} \frac{d^2R}{d\sigma_x^2} \Big|_{\sigma_{x,0}} (\Delta\sigma_x)^2. \quad (1)$$

$\sigma_{x,0}$ is the reference value of the cross section.

The first-order sensitivity coefficient of nuclear data or any other system parameter is expressed as

$$S = \frac{1}{R_0} \frac{\Delta R_1}{p_x} = \frac{1}{R_0} \frac{R_1 p_x}{p_x} = \frac{R_1}{R_0}. \quad (2)$$

Where, ΔR_1 is the first-order Taylor series term and R_1 is the first-order Taylor coefficient. p_x is the relative cross-section perturbation ($p_x \equiv \Delta\sigma_x/\sigma_{x,0}$). R_0 is the unperturbed tally response. In MCNP6, the derivatives appearing in Eq. (1) are evaluated through the PERT card implementation based on a Taylor-series expansion of the response with respect to a relative perturbation p_x . The code internally estimates the first- and second-order response changes (ΔR_1 and ΔR_2) corresponding to a user-defined reference perturbation magnitude. The first-order coefficient $R_1 (= \Delta R_1(p_{x,r})/p_{x,r})$ is obtained by externally dividing the computed first-order response change ΔR_1 by the imposed perturbation magnitude. The sensitivity coefficient is then determined by normalizing this derivative with respect to the unperturbed response.

2.1. Single-response sensitivity coefficient in MCNP

Besides the flux sensitivities to the cross sections, reaction rate sensitivity coefficients can be also determined using MCNP6. Reaction rate calculations are performed using FM card (FMn C m r) specifying the material (m), reaction cross section (r) and the tally multiplier constant (C).

Users must be aware of a caution regarding the sign of the multiplier coefficient (C) when calculating reaction rate sensitivity coefficients. Before addressing this, it is important to understand that the Taylor series coefficients include correction term R_{ij}' (corresponding is 'direct term' in case of SERPENT2) (see ref [2] for more details). The correction term R_{ij}' arises when a tally explicitly depends on the perturbed parameter. That is if a tally in a cell is dependent on a cross section that is perturbed, then the Taylor series coefficients that have a direct-term (correction-term) contribution ($R_{ij}' \neq 0$). R_{ij}' is the fraction of the reaction rate tally involved in the perturbation.

$R_{ij}' \neq 0$ for F4 tallies with FM cards with negative multiplicative constants. If the detector tally uses fission cross section on FM card but capture cross section on RXN is perturbed, then $R_{ij}'=0$. Generally, $R_{ij}'=0$ for F1, F2, F4 tallies without FM cards, and F4 tallies with FM cards with positive multiplicative constants. If $C > 0$, the cross sections are not macroscopic; it is assumed that there is no tally dependence on a perturbed cross section.

Figure 1 gives PERT and FM card examples in case of $R_{ij}' \neq 0$ with PERT4 and R_{ij}' with PERT14. For example, the PERT14 card illustrates the case where $R_{ij}'=0$ and produces the first-order Taylor series term ΔR_1 for changes in the fission reaction rate due to a small perturbation, assuming a linear (first-order) approximation in the (n,γ) cross section.

```
m44 92238 1
f4:n 1
fm4 -1 44 -6
pert4:n cell=1 rho=0.047 mat=4 method=2 rxn=-6

f14:n 1
fm14 1 44 -6
pert4:n cell=1 rho=0.047 mat=4 method=2 rxn=102
```

Figure 1. PERT card examples for $R_{ij} \neq 0$ and $R_{ij} = 0$ cases with two different PERT and FM cards.

3. METHODOLOGY in SERPENT2

When perturbation calculations are enabled, SERPENT2 uses a first-order GPT (Generalized Perturbation Theory) based on the weight correction method. The perturbation formalism follows the established framework described in Refs. [3,4] and is briefly summarized here for completeness. The first step of the practical implementation of the proposed method consists in artificially increasing all the cross-sections of the problem involved in the GPT [4].

$$\Sigma_x = f \Sigma_{x,0} \quad , (for \text{ all cross sections}) \quad (3)$$

This is the key step, as SERPENT2 inherently applies the same relative perturbation to all cross sections. Scaling all cross sections by a factor f simply increases the number of proposed collisions. SERPENT2 applies an accept/reject game to preserve unbiased physics during the particle transport with increased number of events. Then, whenever reaction x is sampled in the neutron tracking routines, collisions are rejected with a probability of $(1 - \frac{1}{f})$. If a value of 2 is adopted for the factor f , reactions are accepted (labeled ACC) and rejected (labeled as REJ) with an equal probability. This correction by f preserves the unbiased physics for the accepted events, while the rejected ones are virtual events (or x reactions that would happen in the perturbed physics) used for sensitivity.

Figure 2 illustrates where ACC/REJ events are placed when all cross sections are perturbed by the same factor along the path of a neutron in one history. For sensitivity analyses with respect to a specific cross section x , only the ACC/REJ event for reaction x is counted. The accepted events are the real (analog/unbiased) collisions occurred in the history of particles. Rejected events are virtual and do not change or enter the real path of particle history. These rejected events (REJ) represent virtual collisions associated with perturbed physics. It means if x cross section was a bit larger, x reaction would happen more often here instead of the non- x reaction that occurred.

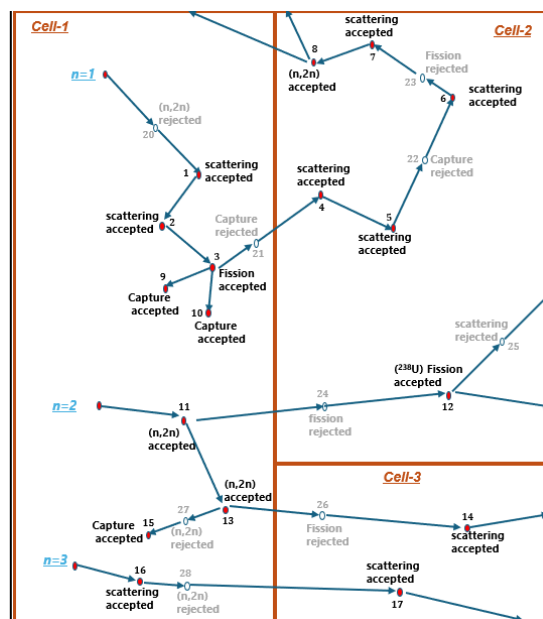


Figure 2: Illustrative collision histories of three source particles with more sampled (proposed) events.

3.1. Weight perturbation (analytically)

Despite the title of ‘weight perturbation’, SERPENT2 does not perturb the weight. The perturbed weight appears in the math of the estimator (i.e., analytical), not as a perturbed weight that would change the particle’s path in the transport [5]. This is a kind of mathematical trick to get the first derivative (sensitivity) from one analog calculation.

If accepted and rejected collisions are both scored, the effect of nuclear data perturbations on the particle weight can be studied by adopting eq. Eq. 4, which is reproduced from Ref. [4] for completeness. Positive perturbations of the cross-section Σ_x for the reaction x will increase the weight of particles that have in their history accepted x collisions. At the same time, when virtual collisions x are present in particle history, the perturbed weight will be reduced.

$$w_n^* \cong w_{n,0} \left(1 + \frac{d\Sigma_{n,2n}}{\Sigma_{n,2n}}\right) \left(1 + \frac{d\Sigma_s}{\Sigma_s}\right) \left(1 - \frac{d\Sigma_f}{\Sigma_f}\right) \left(1 + \frac{d\Sigma_s}{\Sigma_s}\right) \left(1 - \frac{d\Sigma_c}{\Sigma_c}\right) \dots$$

$$\left(1 + \frac{d\Sigma_f}{\Sigma_f}\right) \left(1 + \frac{d\Sigma_s}{\Sigma_s}\right) \left(1 - \frac{d\Sigma_c}{\Sigma_c}\right) \left(1 + \frac{d\Sigma_f}{\Sigma_f}\right) \dots$$

Eq. 4 expresses the perturbed weight as a product over many possible reaction channels, each multiplied by its corresponding relative perturbation factor. In practice, when deriving the sensitivity coefficient with respect to a single reaction perturbation, only the relative change of the weight for that parameter is needed to be considered. As an illustration, suppose that in one history there are two accepted events of reaction x and three rejected scattering of the same type. The product of the corresponding perturbation factors becomes

$$w_n = w_{n,0} \left(1 + \frac{\Delta\Sigma_x}{\Sigma_x}\right) \left(1 + \frac{\Delta\Sigma_x}{\Sigma_x}\right) \left(1 - \frac{\Delta\Sigma_x}{\Sigma_x}\right) \left(1 - \frac{\Delta\Sigma_x}{\Sigma_x}\right) \left(1 - \frac{\Delta\Sigma_x}{\Sigma_x}\right). \quad (5)$$

The relative change in weight due to accepted (a) and rejected (r) event counts is

$$\frac{w_n}{w_{n,0}} = 1 + (a - r)p_x + \mathcal{O}(p_x^2) \quad (6)$$

where, $\mathcal{O}(p_x^2)$ is all the terms of order p_x^2 and higher in the expansion,

First- and second-order Taylor expansions to approximate the perturbed weight due to small perturbations can be also written as below.

$$w_n(p_x) = w_n(p_x = 0) + \frac{dw_n}{dp_x} \bigg|_{p_{x,0}} p_x + \frac{1}{2} \frac{d^2 w_n}{dp_x^2} \bigg|_{p_{x,0}} (p_x)^2 \quad (7)$$

The first derivative term of the Taylor expansion can be written as

$$\frac{1}{w_{n,0}} \frac{dw_n}{dp_x} = (a - r) \quad (8)$$

Since SERPENT2 organizes a history n into generations, eq. 8 is expressed as below using accepted and rejected x reaction events in generation g of history n . Eq. 9 presented in Ref. [4] is the written as:

$$N_{x,n} = \frac{dw_n/w_{n,0}}{dp_x} = \sum_{g=(\alpha-\lambda)}^{\alpha} ((n,g)ACC_x - (n,g)REJ_x) \quad (9)$$

Where, $\frac{dw_n/w_{n,0}}{dp_x}$ or $\frac{dw_n/w_n}{dp_x}$: the derivative of the relative particle weight with respect to a relative perturbation of reaction type x (e.g., capture cross section).



$(n, g)ACC_x$: number of accepted reaction type x in history n , generation g ,

$(n, g)REJ_x$: number of rejected proposals of reaction type x (virtual collisions) in the same history/generation,

$\Sigma_{g=(\alpha-\lambda)}^\alpha$: Sum over (accepted - rejected) events of reaction type x for this history n , but only for the current generation α that contributes to score of detectors and λ , which is the ancestor depth used to propagate the perturbation.

3.2. Single-Response Sensitivity Extraction in SERPENT2

The reaction rate (R) response is expressed as

$$R = q \frac{1}{N \cdot V} \int \phi \Sigma_d dE \quad (10)$$

where, ϕ is flux, Σ_d is the generic space and energy cross section of detector reaction d . q is source intensity, N is population normalization factor, and V is volume, which are omitted in the following as they cancel out in the final sensitivity coefficient formula.

Differentiate both sides of eq. 10 with respect to the relative perturbation p_x of reaction x ($\Sigma_x = \Sigma_{x,0} + \Sigma_{x,0}p_x$).

$$\frac{dR}{dp_x} = \sum_{t \in \alpha} \sum_{n \in t} \frac{\partial}{\partial p_x} (\phi_n \Sigma_d) = \underbrace{\sum_{t \in \alpha} \sum_{n \in t} \frac{d\phi_n}{dp_x} (\Sigma_d)}_{\text{Indirect}} + \underbrace{\sum_{t \in \alpha} \sum_{n \in t} \frac{d\Sigma_d}{dp_x} (\phi_n)}_{\text{Direct (only if } d=x \text{)}} \quad (11)$$

Here, n denotes particle histories, t represents the track segments within each history used in the track-length estimator formulation, and α refers to the particle generation index associated with the detector-contributing history. Eq. 11 has both direct and indirect contributions to the response. As can be seen, the indirect term comes only from changes in the flux/weights (tracked via the accept-reject game). It exists regardless of $d=x$ (e.g., the detector tally uses fission cross section (i.e., $\Sigma_d = \Sigma_{(n,f)}$) but capture cross section (i.e., $\Sigma_x = \Sigma_{(n,g)}$) is perturbed). The direct terms come from the explicit dependence of the tally response on the perturbed parameter (Σ_x).

SERPENT2 does not currently support calculating sensitivity coefficients for a single detector response directly. Instead, the response must be defined as a ratio, such as a ratio of fluxes, reaction rates (e.g., sensitivities to the spectral index of $^{238}\text{U}/^{235}\text{U}$), or ratio of reaction rate to flux (e.g., for microscopic cross section sensitivities). In the end, Eq. 12 is used in SERPENT2 to determine the sensitivity coefficients for the numerator (det.1) and denominator (det.2) detectors. $T_{\Sigma_d, n}$ are tally contribution per history to the detector responses. $R_{1,0}$ and $R_{2,0}$ are unperturbed responses for the numerator and denominator detectors, respectively.

$$S_x^R = \underbrace{\left(\frac{T_{\Sigma_d, n} N_{x, n}}{R_{1,0}} \right)_{\text{det.1}}}_{\text{Indirect, det. 1}} - \underbrace{\left(\frac{T_{\Sigma_d, n} N_{x, n}}{R_{2,0}} \right)_{\text{det.2}}}_{\text{Indirect, det. 2}} + \underbrace{((S_x^R|_{\text{dir.}})_{\text{det.1}})}_{\text{Direct, det. 1}} - \underbrace{(S_x^R|_{\text{dir.}})_{\text{det.2}}}_{\text{Direct, det. 2}} \quad (12)$$

As it is seen, Eq. 12 does not directly compute the ratio of detector 1 to detector 2, but rather combines their respective contributions separately based on the perturbation applied. However, in the SERPENT2 input file, both numerator and denominator detector responses are explicitly required. To overcome this intrinsic ratio-based restriction in SERPENT2, the following practical procedure is adopted. By placing the second detector (det.2) in a region that does not score any physical contributions (e.g., an external world), its nominal value becomes zero. In this configuration, the terms

involving detector 2 vanish in the final sensitivity expression. Consequently, the resulting coefficient reflects only the sensitivity of the numerator detector (det.1). This is the approach adopted in the present SERPENT2 calculations.

4. CALCULATION MODEL

In this example, the objective is to determine the sensitivity coefficients of one reaction rate in the detector region due to nuclear data perturbations. The geometry consists of concentric spherical shells, which start with a central spherical air-filled source region followed by layers of graphite, steel, and lead. A thin outer steel shell encloses the structure, as shown in Figure 3. The detector region (cell-1001) is modeled as a 1 cm thick spherical shell forming the outermost part of the lead layer.

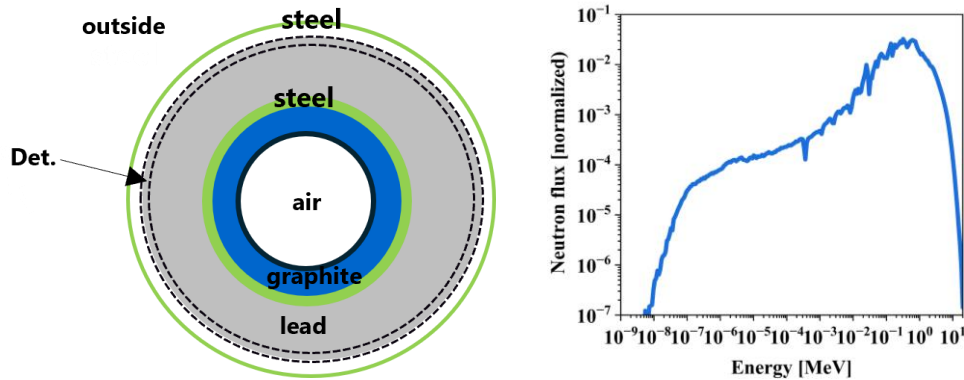


Figure 3. Illustrative concentric spherical geometry (left) and source spectrum (right).

This configuration is inspired by the VENUS-F (CC14) core design. The source spectrum (see Figure 3-right) is preliminarily obtained using the original CC14 model at the inner boundary of the graphite moderator. Since the actual reactor core is not modeled in this spherical approximation, the core region is replaced with air, and neutrons are emitted isotopically from the surface of the inner air sphere.

It is important to note that histogram distributions were used for the source. MCNP6 accepts raw flux intensities via the SP card on the SDEF source definition and performs internal normalization. In contrast, SERPENT2 requires per-bin normalized intensities for each energy bin, specified using the SB card in the SRC block. Notably, this requirement is not explicitly documented in the SERPENT2 manuals.

The goal here is to evaluate how the reaction rate of $^{238}\text{U}(n,f)$ calculated in the detector region is sensitive to the lead cross sections. SENS cards defined in SERPENT2 code are shown in

Figure 4 as an example. To determine the sensitivity to the lead cross sections, the 252-group energy structure from SCALE is used for the perturbation. The energy range of interest for the detector tally is between 1-3 MeV, as the $^{238}\text{U}(n,f)$ reaction rate is dominated by neutrons within this energy interval. Denominator detector (i.e., RR-U235_fis) is in the cell-999, which is located outside the geometry (i.e., “universe” region), therefore, no particles are tracked there, and the resulting scores are zero. Therefore, the obtained sensitivity coefficients represent only the contribution from the numerator detector (i.e., RR-U238_fis).

```

-----
cell cell-999 0 outside sur-200 % Outside world
cell cell-1001 0 m4 -sur-104 sur-109 % detector cell
-----
det RR-U238_fis n de det-reaction-rate-egrid dc cell-1001 dr 18 mat-92238 % numerator
det RR-U235_fis n de det-reaction-rate-egrid dc cell-999 dr 18 mat-92235 % denominator
-----
sens pert matlist m4 % perturbed material
sens pert zaalist 822070 % perturbed ZAI
sens pert xs realist inl % perturbed reaction
-----
sens resp detratio sens_ratio RR-U238_fis RR-U235_fis
-----

```

Figure 4. SENS card used to calculate the sensitivity coefficients.

5. RESULTS

Figure 5 presents the $^{238}\text{U}(n,f)$ reaction rate as a function of energy (top-left), and the sensitivity coefficients of the $^{238}\text{U}(n,f)$ reaction rate with respect to the $^{206}\text{Pb}(n,g)$ (mt=102, top-right), $^{207}\text{Pb}(n,inel)$ (mt=4, bottom-left) and $^{208}\text{Pb}(n,el)$ (mt=2, bottom-right) cross sections, as obtained using MCNP6 and SERPENT2.

Since the total $^{238}\text{U}(n,f)$ reaction rate is dominated by neutrons in the 1–3 MeV energy range, the sensitivity coefficients were calculated using the 1–3 MeV detector energy bin. The sensitivity profiles depend on the selected detector energy bin, and using the dominant energy range ensures that the reported coefficients represent the main contribution to the reaction rate.

Overall, the sensitivity coefficients obtained with both codes show very good agreement across the energy spectrum. This confirms the applicability of the practical method adopted in SERPENT2 for single-response sensitivity calculations and demonstrates its consistency with the MCNP6 perturbation approach.

This validated approach will be applied in future simulations of the full VENUS-F reactor model.

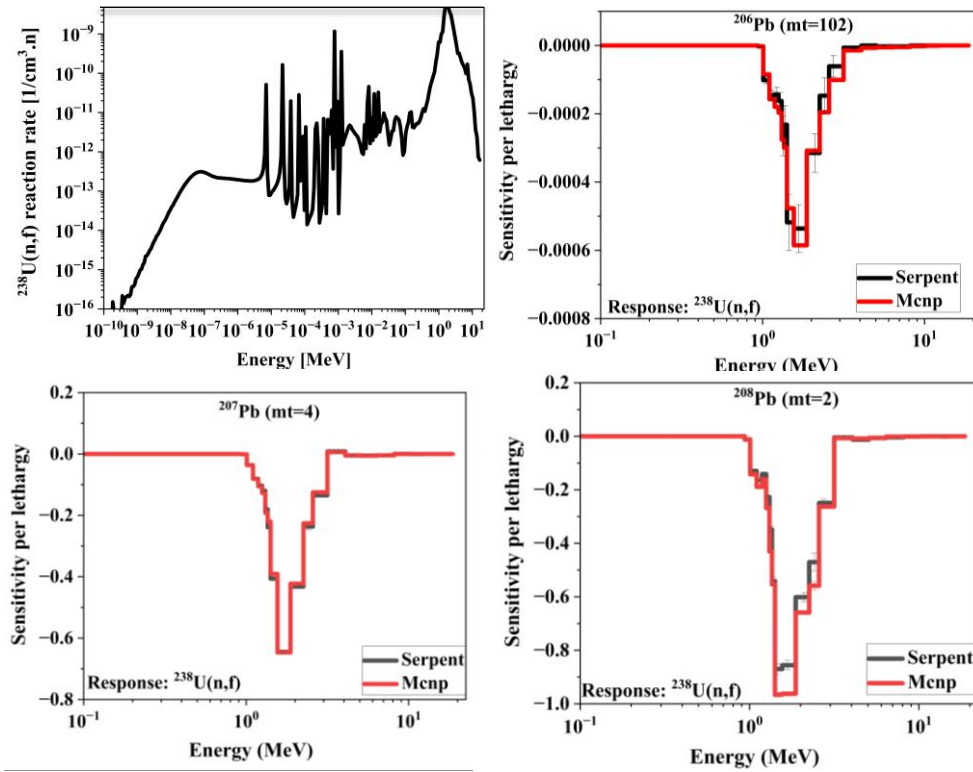


Figure 5. Reaction rate spectrum (top-left) and sensitivity coefficients of $^{238}\text{U}(n,f)$ reaction rate to ^{206}Pb (mt=102, top-right), ^{207}Pb (mt=4, bottom-left) and ^{208}Pb (mt=2, bottom-right) cross sections.

Uncertainty propagation for the $^{238}\text{U}(n,f)$ reaction rate was also performed using sensitivity coefficients from SERPENT2 together with covariance matrices processed with NJOY from the ENDF/B-VIII.0 library, the same evaluation used in the transport calculations. The total variance was obtained using the standard first-order perturbation expression:

$$\sigma_{\text{ND}}^2 = S^T \cdot \text{COV} \cdot S \quad (13)$$

where S is the vector of sensitivity coefficients for the reactions considered and COV is the corresponding covariance matrix.

The ranking of contributors to the total variance is shown in Table I. The dominant source is the $^{207}\text{Pb}(n,\text{inl})$ cross section, followed by $^{208}\text{Pb}(n,\text{el})$. The $^{206}\text{Pb}(n,\gamma)$ contribution is negligible, consistent with its near-zero sensitivity in this configuration. The resulting total relative uncertainty in the $^{238}\text{U}(n,\text{f})$ reaction rate is 3.6%.

Table I. Ranked variance contributions to the propagated $^{238}\text{U}(n,\text{f})$ reaction-rate uncertainty.

Rank	Nuclide	MT	Variance Contribution (%)
1	^{207}Pb	4	72.96
2	^{208}Pb	2	27.04
3	^{206}Pb	102	<0.01

Further investigations will explore the sensitivity of different reaction rate responses across various detector energy bins. The impact of variance reduction techniques on sensitivity coefficients, particularly those introduced by the ADVANTG-generated weight windows, which have been shown to cause significant discrepancies [6], will also be assessed.

6. CONCLUSIONS

This study demonstrates the feasibility of calculating first-order sensitivity coefficients for individual reaction rates in fixed-source problems using both MCNP6 and SERPENT2. While SERPENT2 inherently restricts sensitivities to ratios of detector responses, a validated practical approach was demonstrated to extract single-response sensitivity coefficients. Despite their differing mathematical approaches, comparable results were achieved, which confirms the consistency of the adopted approach.

The results show good agreement between the codes for the principal lead isotopes, such as $^{206}\text{Pb}(n,\text{g})$, $^{207}\text{Pb}(n,\text{inl})$ and $^{208}\text{Pb}(n,\text{el})$, which further support the reliability of the proposed extraction procedure for sensitivity analysis. Future work will extend this methodology to additional detector responses and alternative energy bins. Further investigations will also examine the impact of variance-reduction techniques, including ADVANTG-generated weight windows.

REFERENCES

- [1] F. Grimaldi et al. "The CoRREx neutron spectrum filtering campaign at VENUS-F for calculation-to-experiment discrepancy interpretation." *Annals of Nuclear Energy*, 219, p. 111425 (2025).
- [2] C. J. Werner (editor), "MCNP Users Manual - Code Version 6.2", LA-UR-17-29981 (2017)
- [3] J. Favorite, Using the MCNP Taylor series perturbation feature (efficiently) for shielding problems, ICRS-13 & RPSD-2016, EPJ Web of Conferences 153, 06030 (2017), DOI: 10.1051/epjconf/201715306030.
- [4] Manuele Aufiero, et al., A collision history-based approach to sensitivity/perturbation calculations in the continuous energy Monte Carlo code SERPENT, *Annals of Nuclear Energy* 85 (2015) 245–258.
- [5] V. Valtavirta, et al., Collision history based sensitivity/ perturbation calculation capabilities in SERPENT 2.1.30, ANS Best Estimate Plus Uncertainty International Conference (BEPU 2018), BEPU2018-240, Italy.
- [6] Juan A. Monleon de la Lluvia, et al., Toward Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations, *Nuclear Science and Engineering* (2025), DOI: 10.1080/00295639.2025.2510048